

HEESEUNG SHON, Ph.D.

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Work Authorization: U.S. Permanent Resident (*No Sponsorship required)

RESEARCH INTERESTS

Transportation systems analysis, planning, and operations
Transportation asset management and intelligent infrastructure
Data science, AI/ML, and simulation for transportation decision support
Infrastructure safety, monitoring, and performance evaluation
Optimization and decision-support models for mobility and infrastructure systems

EDUCATION

Korea Advanced Institute of Science and Technology (KAIST), Daejeon, Korea

Ph.D. in the Graduate School of Mobility, 2023

Advisor: Prof. Jinwoo Lee

Dissertation: Optimal road asset management system considering condition inspection policies and information value

Korea University, Seoul, Korea

M.S. in Civil and Environmental Engineering, 2019

Advisor: Prof. Seungmo Kang

Thesis: New section based congestion index for applications in enhancing missing value imputation and predicting travel time

Korea University, Seoul, Korea

B.S. in Civil and Environmental Engineering, 2017

ACADEMIC EXPERIENCE

University of Illinois Chicago, Illinois, United States

Postdoctoral Researcher, 2024-2025

Visiting Researcher, 2023-2024

KAIST, Daejeon, Korea

Postdoctoral Researcher, 2023

Researcher, 2019-2020

JOURNAL PUBLICATIONS

Transportation Asset Management

- **Shon, H.** and Lee, J., 2021. Integrating multi-scale inspection, maintenance, rehabilitation, and reconstruction decisions into system-level pavement management systems. *Transportation Research Part C: Emerging Technologies*, 131, p.103328.
- **Shon, H.**, Cho, C., Byon, Y. and Lee, J., 2022. Autonomous condition monitoring-based pavement management system (ACM-PMS). *Automation in Construction*, 138, p.104222.

Transportation Systems Planning and Operations

- Lee, J., **Shon, H.**, Papakonstantinou, I. and Son, S., 2021. Optimal fleet, battery, and charging infrastructure planning for reliable electric bus operations. *Transportation Research Part D: Transport and Environment*, 100, p.103066.
- Jeong, M., **Shon, H.**, Kim, J., Jung, Y. and Lee, J., 2023. Analysis of traffic load reduction effects of overloading enforcement system from a traffic assignment perspective, *International Journal of Highway Engineering*, 25(6), pp.309-315.
- **Shon, H.**, Kim, S., and Lee, J., 2024. Optimal planning of urban air mobility systems accounting for ground access trips. *International Journal of Sustainable Transportation*, 18(4), pp.356-378.
- **Shon, H.** and Lee, J., 2025. An optimization framework for urban air mobility (UAM) planning and operations. *Journal of Air Transport Management*, 124, p.102720.
- Perez, D., **Shon, H.**, Zou, B. and Kuhn, K., 2025. Advanced air mobility for commuting? An exploration of economic, energy, and environmental feasibility. *Transport Economics and Management*, 3, pp.135-152.

Big Data Analysis and Deep Learning

- Choi, Y.Y., **Shon, H.**, Byon, Y.J., Kim, D.K. and Kang, S., 2019. Enhanced application of principal component analysis in machine learning for imputation of missing traffic data. *Applied Sciences*, 9(10), p.2149.
- **Shon, H.**, Kim, S., Yoon, Y. and Lee, J., 2026. Incorporating Traffic Flow Transmission Impacts into Dynamic Graph Diffusion Convolutional Network for Traffic Speed Prediction. *Journal of Intelligent Transportation Systems: Technology, Planning, and Operations*, Accepted.

PEER-REVIEWED CONFERENCE PROCEEDINGS

- **Shon, H.** and Lee, J. 2022. Optimal Urban Air Mobility (UAM) Planning and Operations. *Transportation Research Board (TRB) 102nd Annual Meeting*. Accepted for presentation.
- **Shon, H.**, Kim, S. and Lee, J., 2022. Incorporating Traffic Flow Transmission Impacts into Spatio-Temporal Graph Diffusion Convolutional Network for Traffic Speed Prediction. *Transportation Research Board (TRB) 102nd Annual Meeting*. Accepted for presentation.
- Selmoune, A., **Shon, H.**, Han, J., Kim, J. and Lee, J., 2022. Heterogeneity in Bus Choice Behavior by Time of Day: Empirical Evidence from Smart Card Big Data. *Transportation Research Board (TRB) 102nd Annual Meeting*. Accepted for presentation.
- **Shon, H.**, Cho, C., Byon, Y. and Lee, J., 2022. Initial Concept of Autonomously Monitored Pavement Management Systems and Its Potential Benefits. *Transportation Research Board 101st Annual Meeting*. Accepted for presentation.
- **Shon, H.** and Lee, J., 2021. Information Value in Pavement Management Systems with Surface Condition Information Collected by Autonomous Vehicles. *The 14th International Conference of Eastern Asia Society for Transportation Studies (EASTS)*.
- **Shon, H.** and Lee, J., 2021. Joint optimization of multi-scale decisions in budget allocation, inspection frequency, and maintenance policies for transportation infrastructure systems. *Transportation Research Board 100th Annual Meeting*. Accepted for presentation.
- Oh, Y., **Shon, H.**, Jeong, H., Bae, HG. and Kang, S., 2018. Analysis on taxi operation using digital tachograph and meter data in Korea. *18th International Conference Road Safety on Five Continents (RS5C 2018)*.

HONORS AND AWARDS

Best paper award, The 86th Conference of Korean Society of Transportation, 2022
 Best paper award, Conference of Korean Society of Civil Engineering, 2021
 Best paper award, The 83rd Conference of Korean Society of Transportation, 2020
 President award, Korean Society of Transportation, 2017

SELECTED RESEARCH & APPLIED PROJECTS

Big-Data- and Optimization-Driven Sustainable Transport Infrastructure Asset Management, *National Research Foundation of Korea*

Developed multi-objective life-cycle maintenance optimization models and proposed an autonomous monitoring-based framework for road pavement management.
 Project outcomes resulted in peer-reviewed journal publications.

Road Asset Management System (RAMS) Strategy for the Dominican Republic, *The Export-Import Bank*

Designed a country-tailored RAMS framework, including data collection and inspection strategies, long-term maintenance policy, and system architecture, and developed phased implementation roadmap.
 Principal author of the technical report "Road Asset Management System Strategy for the Dominican Republic."

Urban Mobility Network Analysis & Forecasting with Attention Mechanisms and Graph Neural Networks, *National Research Foundation of Korea*

Built graph-based deep learning models with attention and diffusion mechanisms to learn taxi-flow spatial correlations and forecast traffic speeds. (Methods/Tools: Gated Recurrent Unit, Diffusion Convolutional Network, Python)

Taxi Operation Information Management System, *Neighbor System*

Analyzed large-scale taxi data to identify demand hotspots and developed prototype detectors for regulatory violations and unauthorized operations. (Methods/Tools: Statistics, Python, GIS)

Jeju Innovation City Smart-Mobility Living Lab, *Jeju Free International City Development Center*

Developed an end-to-end prototype for detecting challenging perception cases using roadside video infrastructure and delivering hazard alerts to autonomous vehicles. (Methods/Tools: Computer vision, Python, video analytics)

SKILLS

Programming & Data Analysis: Python, MATLAB, R, C++

Transportation & Simulation Tools: EMME, VISSIM, CORSIM

Modeling & Analytics: Optimization, simulation, statistical analysis, machine learning, data analytics

Other Tools: LaTeX, Microsoft Office

ADDITIONAL

Languages: English (Fluent), Korean (Native)

Military Service: Republic of Korea Army, Sergeant